ATUL TANAJI MOHITE

Room 1.04, E 2.6, Saarland University, Saarbruecken, Germany
atul.mohite@uni-saarland.de ⋄ atulmohite1326@gmail.com
Google scholar ⋄ https://atulmohite1326.github.io/webpage/



EDUCATION

Saarland University, Saarbruecken *Nov 2023 - Present*
Doctoral researcher in department of Physics

University of Luxembourg, Luxembourg *Nov 2020 - Nov 2022*
Doctoral researcher in department of Physics

Ludwig Maximilian University, Munich *Oct 2017 - April 2020*
Master of Science in Physics GPA: 1.3

Indian Institute of Technology, Delhi *Jul 2013 - May 2017*
Bachelor of Technology in Engineering Physics DGPA: 7.913 / 10

RESEARCH INTERESTS

Topics: stochastic thermodynamics, non-equilibrium thermodynamics, information geometry, non-linear dynamics, stochastic processes, and soft condensed matter
I am interested in non-linear and non-equilibrium phenomena in physics. This includes many problems in theoretical biophysics and non-equilibrium statistical physics. My main focus is stochastic thermodynamics, coarse-graining, geometry, and the interplay of liquid-liquid phase separation and chemical reactions in biological systems, aimed at quantifying their thermodynamic dissipation constraints.

PUBLICATIONS

A.T. Mohite and H. Rieger, Stochastic thermodynamics of non-reciprocally interacting particles and fields, Phys. Rev. E 113, 064136 (2026).

A.T. Mohite and H. Rieger, Thermodynamically Consistent Coarse-graining: from Interacting Particles to Fields via Second Quantization, arXiv:2508.11430.

A.T. Mohite and H. Rieger, Minimum Action Principle for Entropy Production Rate of Far-From-Equilibrium Systems, arXiv:2511.00967.

A.T. Mohite and H. Rieger, Thermodynamic length in stochastic thermodynamics of far-from-equilibrium systems: Unification of fluctuation relation and thermodynamic uncertainty relation, Phys. Rev. E 114, 014103 (2026).

A.T. Mohite and H. Rieger, Generalized finite-time optimal control framework in stochastic thermodynamics, Phys. Rev. Research (2026).
K. Proesmans, G. Falasco, **A.T. Mohite,** M. Esposito, E. Fodor, Quantifying dissipation in flocking dynamics: When tracking internal states matters, Phys. Rev. E 112, 024103 (2025).

SCHOLASTIC ACHIEVEMENTS

IIT-Joint Entrance Examination 2012-13
Secured All India Rank 1474 in IIT-JEE 2012-13

Indian Physics Olympiad (IPhO)
Selected in the top 300 students in India, 2012-13

Middle-school Aptitude and Maths test, 2007-08
Awarded 17th rank in Maharashtra state of India with a percentile score of 99.99, 2007-08.

WORK EXPERIENCE

Saarland University, Saarbruecken *Nov 2023 - Oct 2026*
Doctoral researcher in the field of Theoretical Statistical Physics.

University of Luxembourg, Luxembourg *Nov 2020 - Nov 2022*
Doctoral researcher in the field of Theoretical Statistical Physics.

Max Planck Institute for Neurobiology, Martiensried *Dec 2018 - Mar 2020*
Contributed to a team of students for maintenance of SyConn. SyConn is aimed to automate synaptic connectivity inference for volume electron microscopy.

CONFERENCES AND WORKSHOPS

Non-Equilibrium Physics of Non-Conservative Forces, Stuttgart	Invited talk: *Sep 2026*
SigmaPhi 2026, Kolymvari Crete	Contributed talk: *Jul 2026*
IDeAS26, MPI-PKS, Dresden	Poster: *May 2026*
DPG, Dresden	Contributed talk: *Mar 2026*
DPG, Regensburg	Contributed talk: *Mar 2025*
DPG, Berlin	*Mar 2024*
DPG, Regensburg	Contributed talk: *Sept 2022*
Cell Physics, Saarbruecken	*Oct 2023*
Third Infinity conference, Goettingen	Contributed talk: *Sept 2022*
DPG, Regensburg	Contributed talk: *Sept 2022*
WOST III Workshop, Online	Poster: *May 2022*
Journees de Physique Statistique, Paris	Contributed talk: *Jan 2022*
Inhomogeneous Random Systems, Paris	Contributed talk: *Jan 2022*
WOST II Workshop, Online	*May 2021*
CeNS/CRC235 Workshop Evolving Nanosciences	*Sep 2019*
MECO44, Key Challenges in Statistical Physics, Kloster Seeon	*May 2019*

SUMMER SCHOOLS

Theory of Large Deviations and Applications, Les Houches	*Jul 2024*
Physics of Life Summer School, Edinburgh	Poster: *Apr 2022*
Outstanding Challenges in Nonlinear Dynamics, Les Houches	Poster: *Mar 2022*
International Summer School FPSP XV, Brunneck	Poster: *Jul 2021*
The Beg Rohu Summer School, Quiborn	Poster: *Jun 2021*
Arnold Sommerfeld School - Physics of Life, Munich	*Oct 2019*

TEACHING EXPERIENCE

Saarland University *Nov 2023 - Feb 2024*
Statistical Physics, Winter semester 2023

University of Luxembourg *Feb 2022 - Jul 2022*
Non-equilibrium soft and active matter, Summer semester 2022

University of Luxembourg *Sep 2021 - Dec 2021*
Calculation methods for Physics and Mathematics, Winter semester 2021

Ludwig Maximilian University, Munich *Nov 2018 - Feb 2019*
Calculation methods for Physics and Mathematics, Winter semester 2018

INTERNSHIPS

Leiden University, Netherlands *May 2016 - Jul 2016*
Implemented experimental setup for a tomography technique of NbN superconducting multiphoton detectors and analyzed data for it.

TIFR, Mumbai *May 2015 - Jul 2015*
Conducted experiments and examined polarization dependence of dielectric metastructures

TECHNICAL SKILLS

Programming Languages	C++, Java, Python, OOP, HPC
Modeling and Analysis	COMSOL, Mathematica
Software & Tools	Latex

LANGUAGES

Marathi	Mother tongue
Hindi	Fluent
English	Fluent
German	Proficiency B1

HOBBY AND INTERESTS

As a hobby, I like to paint and sketch. I enjoy reading novels and poetry. I like to play strategy board games like Chess. I admire the works of Bhalachandra Nemade and Tukaram.